QMS-FLOW SPECIFIC REQUIREMENTS

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Video Link: https://youtu.be/yuUnNuxrhx8?si=22w_G5CcI2rzAL3A

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PERSONA 1 (QUALITY MANAGER)

Amina Rahman, a 42-year-old Quality Manager at Company XYZ, oversees all ISO 9001:2015 compliance activities including internal audits, CAPA processes, document control, and management review documentation. She has solid experience with digital systems and prefers structured workflows that reduce administrative workload. Amina is highly motivated to maintain the company's certification status and avoid external audit non-conformities. Her main goals are to gain real-time visibility over QMS performance, ensure departments close findings on time, and maintain an organized central repository of documents. Behaviorally, she checks the system dashboard multiple times a day, assigns tasks through automated reminders, and follows up on overdue items weekly. Her major pain points include scattered audit evidence, inconsistent document versions, difficulty tracking CAPA progress, and the heavy reliance on manual reminders when using spreadsheets and shared drives.

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AS A QUALITY MANAGER, I WANT TO...

- schedule internal audits so that all departments follow the QMS audit cycle.
- assign audit findings to users so that corrective actions are properly tracked.
- control document changes through a structured approval workflow so that all updates to SOPs are properly reviewed and traceable.
- track the investigation progress of each complaint so that I can ensure issues are resolved within the expected timeframe.
- link each CAPA to its root cause so that corrective actions address the underlying problem effectively.
- document the results of effectiveness checks so that future audits can verify that the CAPA actions have worked

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PERSONA 2 (INTERNAL AUDITOR)

Raymond Lim is a 35-year-old Internal Auditor with extensive knowledge of ISO 9001:2015 and its compliance requirements. He performs audits across various departments and prefers using tablets during inspections to log findings immediately. Raymond is motivated by efficiency and accuracy, aiming to complete audits with minimal administrative overhead. His goals include generating standardized checklists, ensuring findings are recorded consistently, and improving traceability of evidence. His behaviours include using templates, uploading photos directly during audits, and performing real-time classification of findings. Raymond's main pain-points stem from manual audit processes, such as version confusion between checklist files, misplacement of findings, and difficulty consolidating audit data across departments.

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AS AN AUDITOR, I WANT TO...

- generate audit checklists from templates so that I can prepare audits quickly.
- record audit findings directly into the system so that documentation is accurate and centralized.
- upload photos or files as evidence so that findings are properly supported.
- classify each finding so that the Quality Manager can take appropriate action.

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PERSONA 3 (DEPARTMENT HEAD)

Siti Norhidayah, a 39-year-old Production Department Head, manages a large team on the factory floor and is responsible for ensuring her department complies with SOPs, training requirements, and QMS policies. Her motivations are to maintain smooth operations and avoid disciplinary or compliance issues. Her goals include timely closure of assigned CAPAs, ensuring her staff follow the latest SOPs, and keeping training records updated. She often checks the system upon receiving notifications and tends to rely on clear dashboards for quick updates. Behaviourally, she responds promptly when alerted, uses the system mostly through desktop computers, and reviews SOPs before briefing her team. Her pain-points include unclear SOP versioning, difficulty manually tracking employee competency, and missing deadlines due to unstructured manual tracking systems.

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AS A DEPARTMENT HEAD, I WANT TO...

- view all the CAPAs assigned to me so that I can prioritize actions.
- access the latest SOP versions so that my team follows the correct procedures.
- update the status of audit findings so that the Quality Manager is informed of progress.
- check employee training status so that I can ensure competency compliance.
- review and approve document changes so that only accurate and department-validated SOP updates are released

PERSONA 4 (FACTORY OPERATOR-GENERAL EMPLOYEE)

Amirul Hakimi, a 28-year-old factory operator, interacts with the QMS system mainly when he needs to review SOPs or report operational issues. He prefers simple interfaces and minimal steps as he is not highly tech-savvy. Amirul is motivated to do his job safely and correctly, with the goal of always using the correct, most updated procedure. He typically uses the system only occasionally, behaves cautiously when handling digital forms, and relies on straightforward navigation. His pain-points include difficulty locating correct SOP versions, lack of confidence with complex digital forms, and confusion caused by multiple document formats stored manually by supervisors.

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AS AN EMPLOYEE, I WANT TO...

- easily access the latest SOP so that I can follow correct work procedures.
- log a complaint or issue so that management can address it.
- see my training history so that I know if refresher training is required.

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SYSTEM FEATURES

Use Case Diagram

Domain Model

State Machine Diagram

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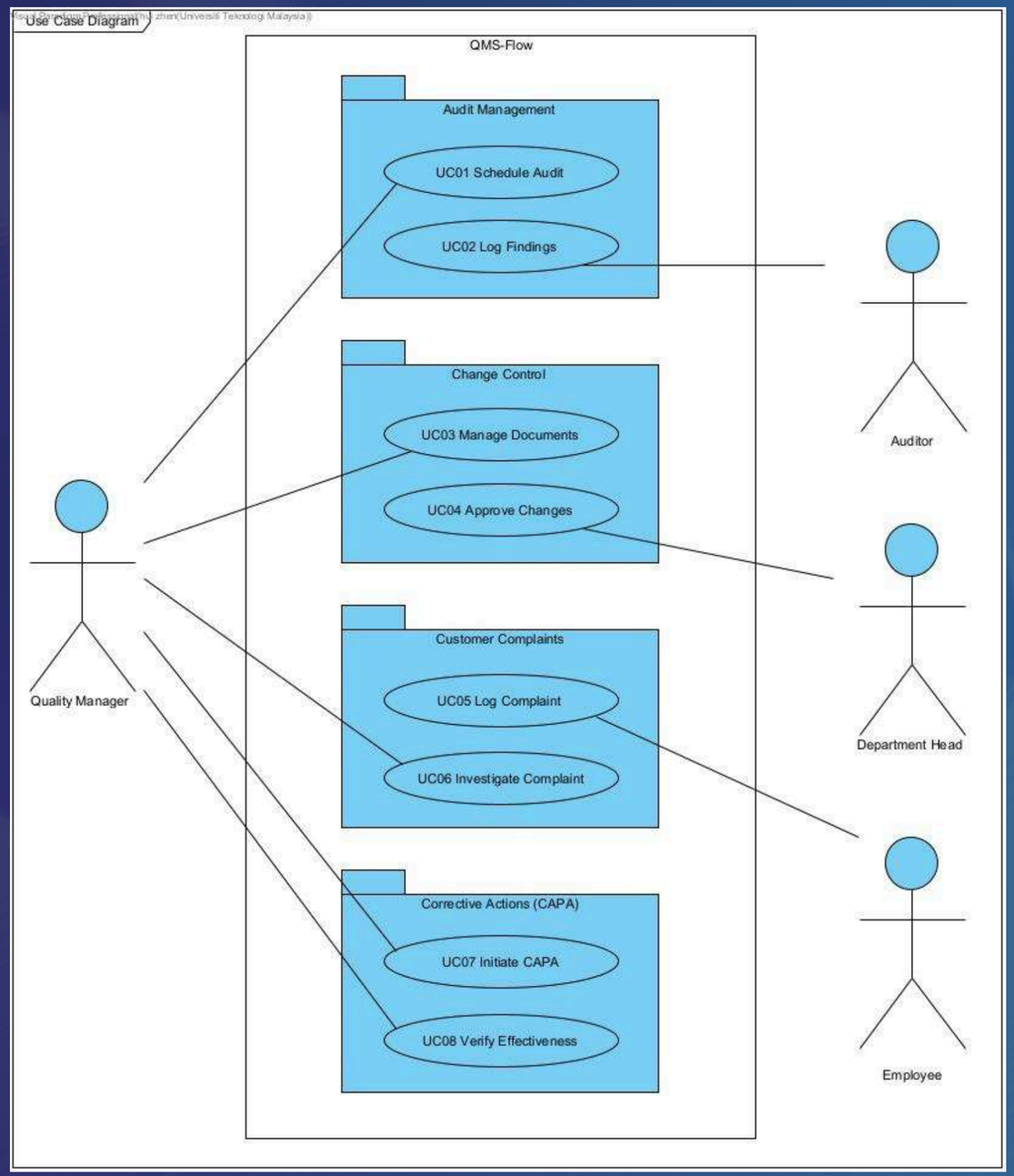

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Use Case ID	Function	Description
UC01	Schedule Audit	This use case allows the Quality Manager to create an annual audit plan, scheduling audits for specific departments and assigning auditors
UC02	Log Findings	This use case allows Auditors to record audit observations or non-conformities directly into the system and assign them to department heads.
UC03	Manage Documents	This use case allows the Quality Manager to upload Standard Operating Procedures (SOPs) to a centralized repository with automated version control (e.g., v1.0, v1.1).
UC04	Approve Changes	This use case enables Department Heads to review and digitally approve document changes. During this process, the document is "locked" (read-only).
UC05	Log Complaint	This use case allows any Employee to log customer feedback or complaints, capturing details such as customer name, product, and nature of the issue.
UC06	Investigate Complaint	This use case allows the Quality Manager to assign a complaint to an investigator and set a target date for resolution.
UC07	Initiate CAPA	This use case allows the Quality Manager to create a Corrective Action (CAPA) record linked to a source (Audit Finding or Customer Complaint) to address root causes.
UC08	Verify Effectiveness	This use case allows the Quality Manager to record the results of effectiveness checks after a corrective action has been implemented.

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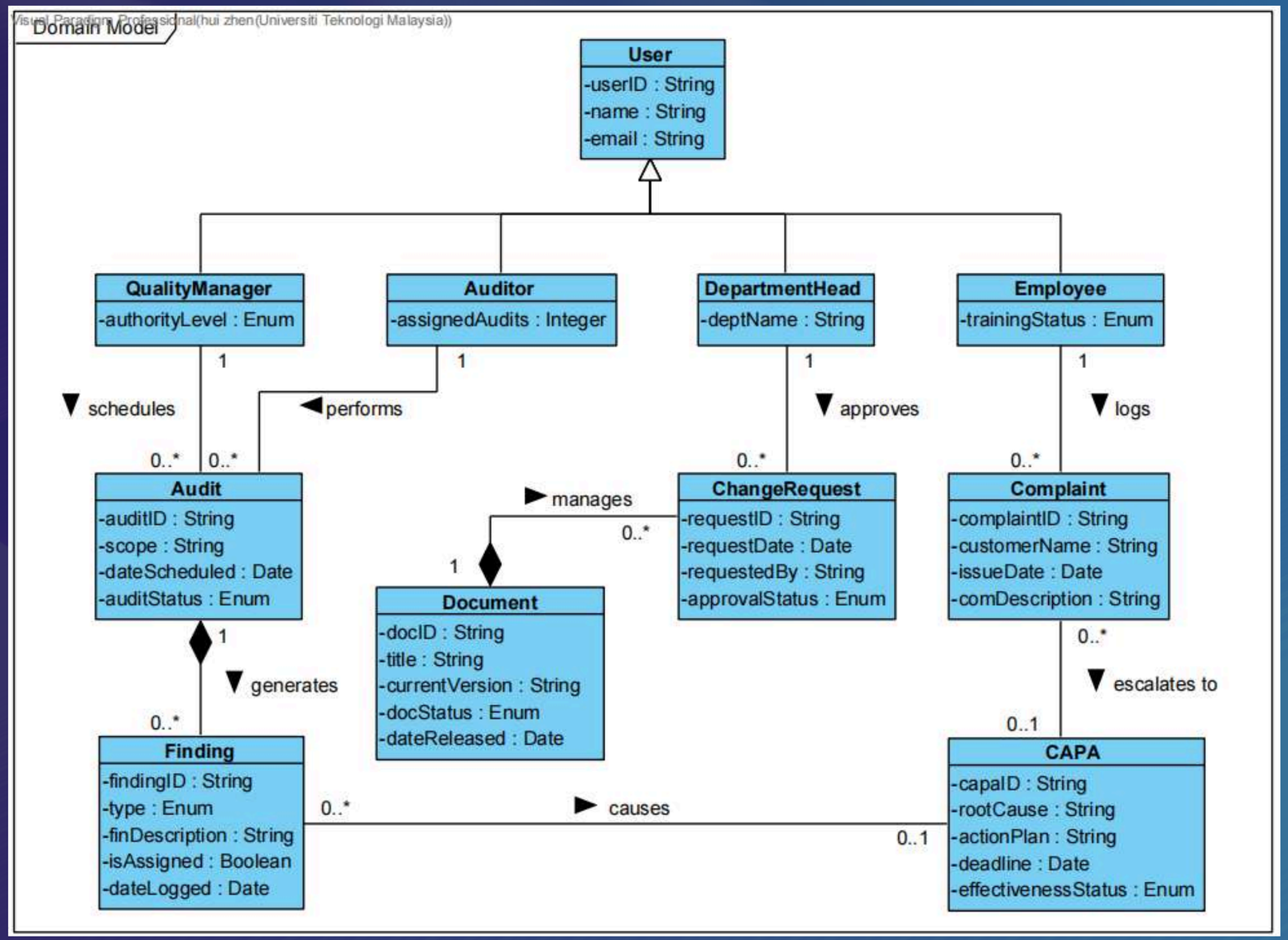

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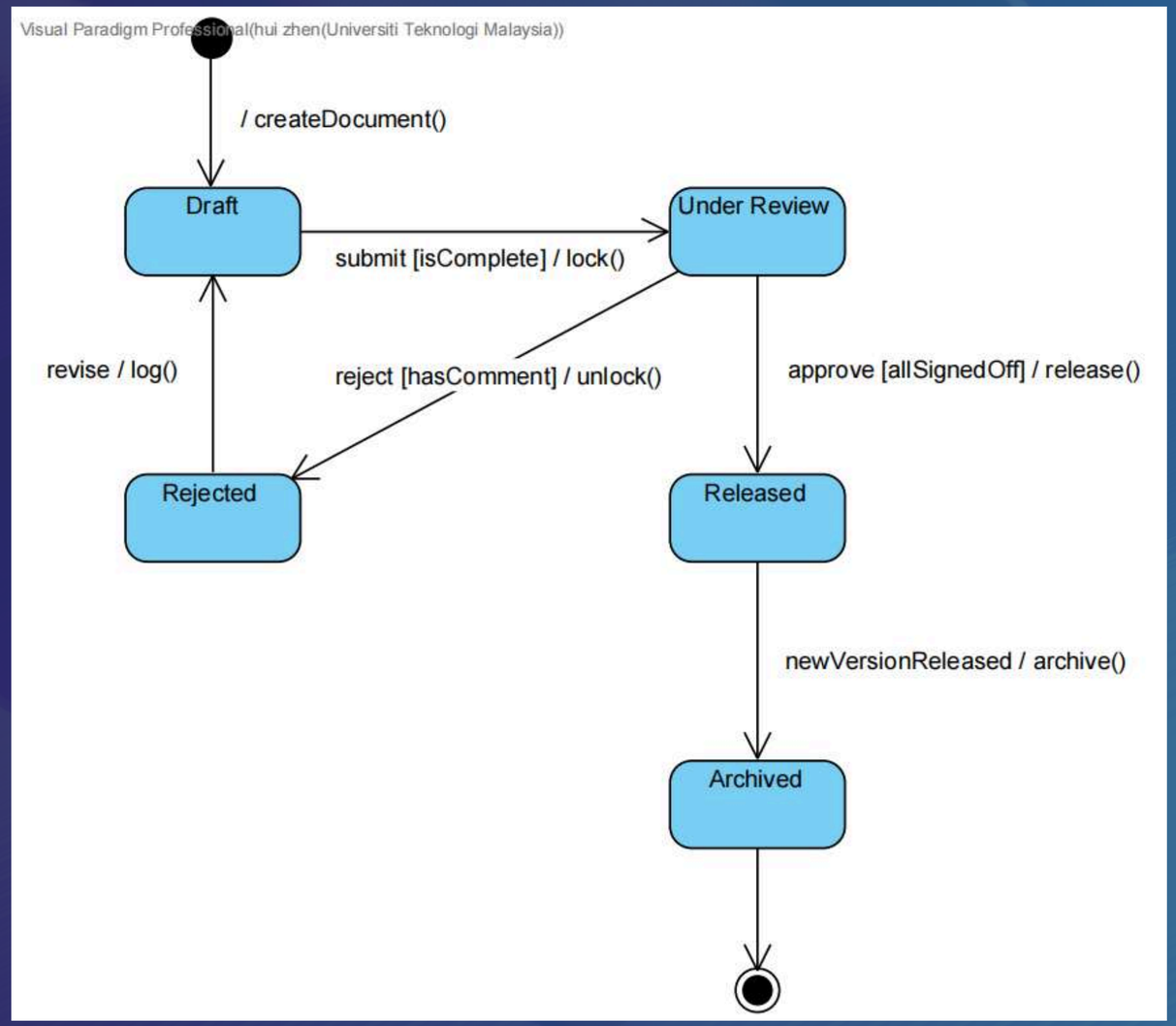

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USER STORY DETAILS

User Story Description

Sequence Diagram

Activity Diagram

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User story: Schedule Audit (UC01)

ID: US001

User Story Description
As a Quality Manager, I want to create an annual audit plan, schedule audits for specific departments, and assign auditors so that the company can maintain a structured audit cycle and ensure ISO 9001:2015 compliance.

Preconditions:

1. The Quality Manager is authenticated and logged into the system.
2. The Quality Manager has the role permission "Quality Manager" or "Audit Manager".
3. The Audit Management module is accessible.

Flow of events:

1. The Quality Manager navigates to the Audit Management dashboard.
2. The system displays the current audit plan and an option to "Schedule New Audit".
3. The Quality Manager clicks "Schedule New Audit".
4. The system displays a form with fields: Audit Title, Department, Scheduled Date, Auditor, and Scope.
5. The Quality Manager fills in the required fields and selects an auditor from a dropdown list of available auditors.
6. The Quality Manager clicks "Save Audit".
7. The system validates the input (e.g., date not in the past, auditor is available).
8. The system saves the audit record and assigns it a unique Audit ID.
9. The system sends a notification to the assigned auditor.
10. The system logs the audit creation in the audit trail.
11. The system displays a success message: "Audit scheduled successfully."
12. Use case ends.

Alternative flow (A1): Auditor Unavailable
If the selected auditor is already assigned to another audit on the same date (Step 7), the system displays an error: "Selected auditor is unavailable on the chosen date. Please select another auditor." The form remains open for correction.

Alternative Flow (A2): Past Date Entered
If the scheduled date is in the past (Step 7), the system displays an error: "Audit date cannot be in the past. Please select a future date."

Postconditions:

1. A new audit record is created with status "Scheduled".
2. The assigned auditor receives a notification.
3. The audit is added to the annual audit plan and visible in the audit calendar.
4. The audit creation is logged in the system audit trail.

Acceptance Criteria

1. The system must prevent scheduling an audit with a past date.
2. The system must check auditor availability and prevent double-booking.
3. All mandatory fields (Audit Title, Department, Date, Auditor) must be validated before saving.
4. The audit must be assigned a unique Audit ID upon creation.
5. The assigned auditor must receive an automated notification (email/system alert).

Exception flow (E1): Database Connection Error
If the system fails to save the audit due to a database error, the transaction is rolled back and an error message is displayed: "System error. Please try again." The form data is retained for re-submission.

Exception flow (E2): Form Timeout
If the session times out during form submission, the system redirects to the login page and all unsaved data is lost. Upon re-login, the user is returned to the Audit Dashboard.

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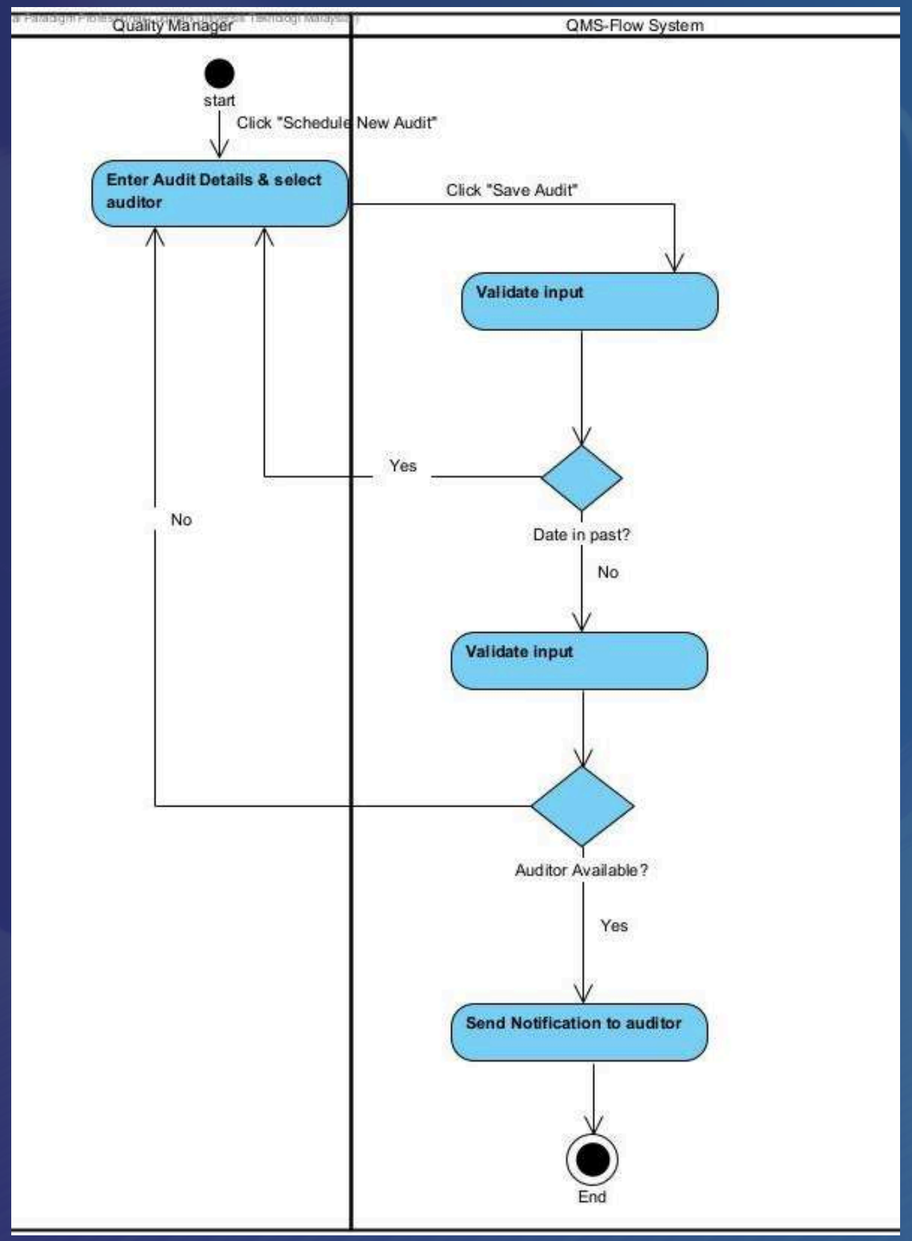

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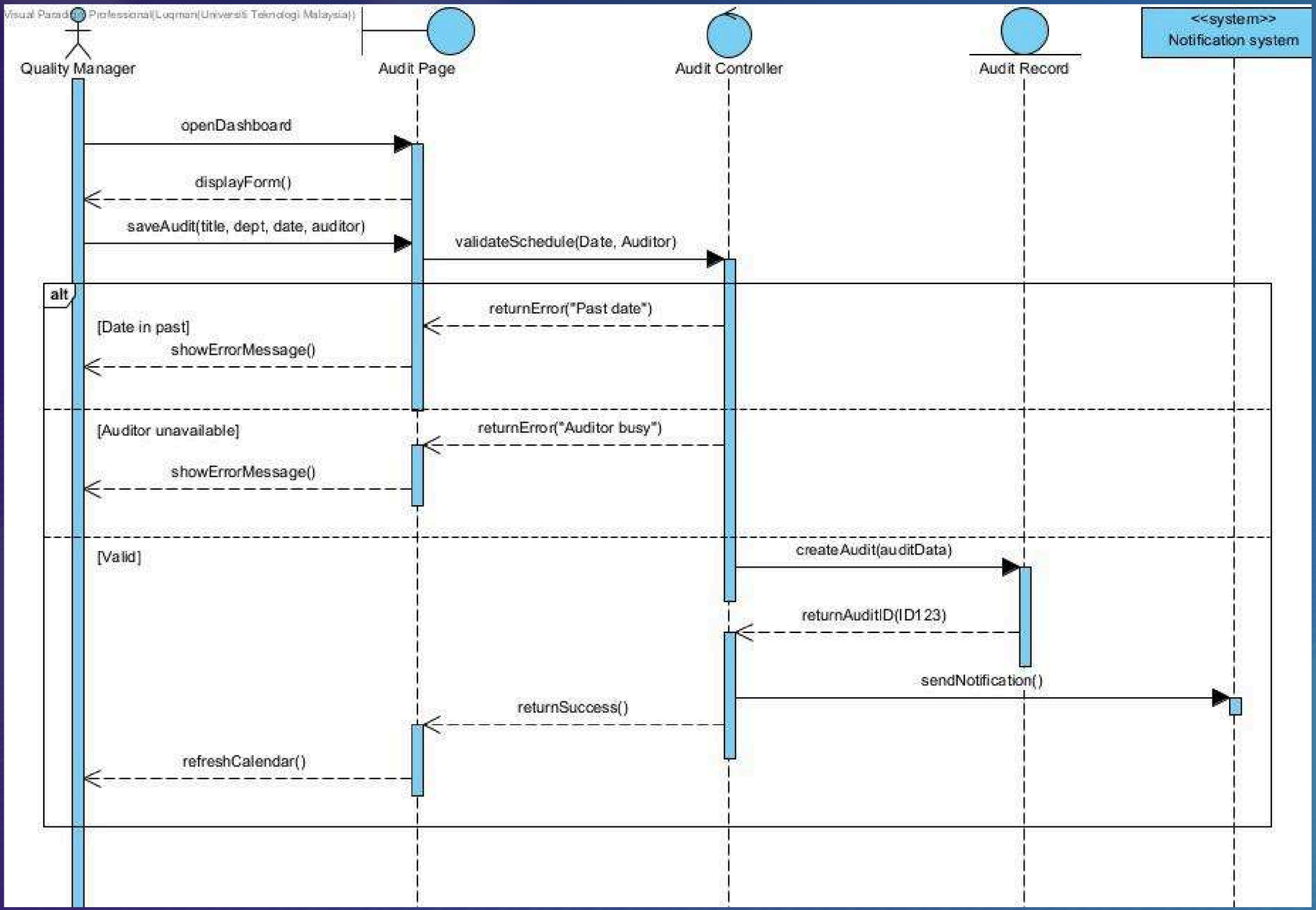

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User story: <Manage Documents> (UC03)
ID: US003
User Story Description As a Quality Manager, I want to upload Standard Operating Procedures (SOPs) to a centralized repository with automated version control so that all users access the latest approved documents and version history is maintained.
Preconditions: 1. Quality Manager is authenticated and logged in. 2. Users have "Document Controller" or "Quality Manager" role permissions. 3. The document repository module is accessible.
Flow of events: 1. Quality Manager navigates to the Document Control module. 2. The system displays the document repository with existing documents and an "Upload New Document" button. 3. Quality Manager clicks "Upload New Document". 4. The system displays an upload form with fields: Document Title, Document Type (SOP, Policy, Work Instruction), Department, Effective Date. 5. Quality Manager selects a file from local storage (PDF, DOCX, XLSX). 6. Quality Manager fills in required metadata and clicks "Submit". 7. System validates the file format (accepts .pdf, .docx, .xlsx, max 50MB). 8. System checks for duplicate document titles. 9. The system generates a new version number (e.g., v1.0) and saves the document. 10. The system logs the upload activity in the audit trail. 11. The system displays a success message. 12. Use case ends.

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Alternative flow (A1): Duplicate Document Title

If a document with the same title already exists (Step 8), system prompts: "Document '[Title]' exists. Do you want to create a new version?" If Yes, the system creates the next version (e.g., v1.1); if No, returns to the upload form.

Alternative flow (A2): Invalid File Format

If the uploaded file is not in accepted format (Step 7), the system displays error: "Invalid file format. Accepted formats: PDF, DOCX, XLSX."

Postconditions:

1. New document is saved in a repository with status "Draft".
2. The document is automatically assigned the appropriate version number.
3. Audit trail records the upload with timestamp and user.
4. Document is locked for editing while in the repository.

Acceptance Criteria

1. The system must accept only .pdf, .docx, and .xlsx file formats.
2. New documents must be assigned version v1.0; updated documents must increment version (v1.0 → v1.1).
3. All required metadata fields must be validated before submission.

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Postconditions:

1. New document is saved in a repository with status "Draft".
2. The document is automatically assigned the appropriate version number.
3. Audit trail records the upload with timestamp and user.
4. Document is locked for editing while in the repository.

Acceptance Criteria

1. The system must accept only .pdf, .docx, and .xlsx file formats.
2. New documents must be assigned version v1.0; updated documents must increment version (v1.0 → v1.1).
3. All required metadata fields must be validated before submission.
4. The system must detect duplicate titles and provide appropriate options.
5. AC5: Upload activity must be logged in the audit trail with user ID and timestamp.

Exception flow (E1): File Too Large

If the uploaded file exceeds 50MB, the system displays an error: "File size exceeds limit. Maximum size: 50MB." and prevents upload.

Exception flow (E2): Database Error

If the system fails to save a document due to database error, the transaction is rolled back and an error message displayed: "System error. Please try again.

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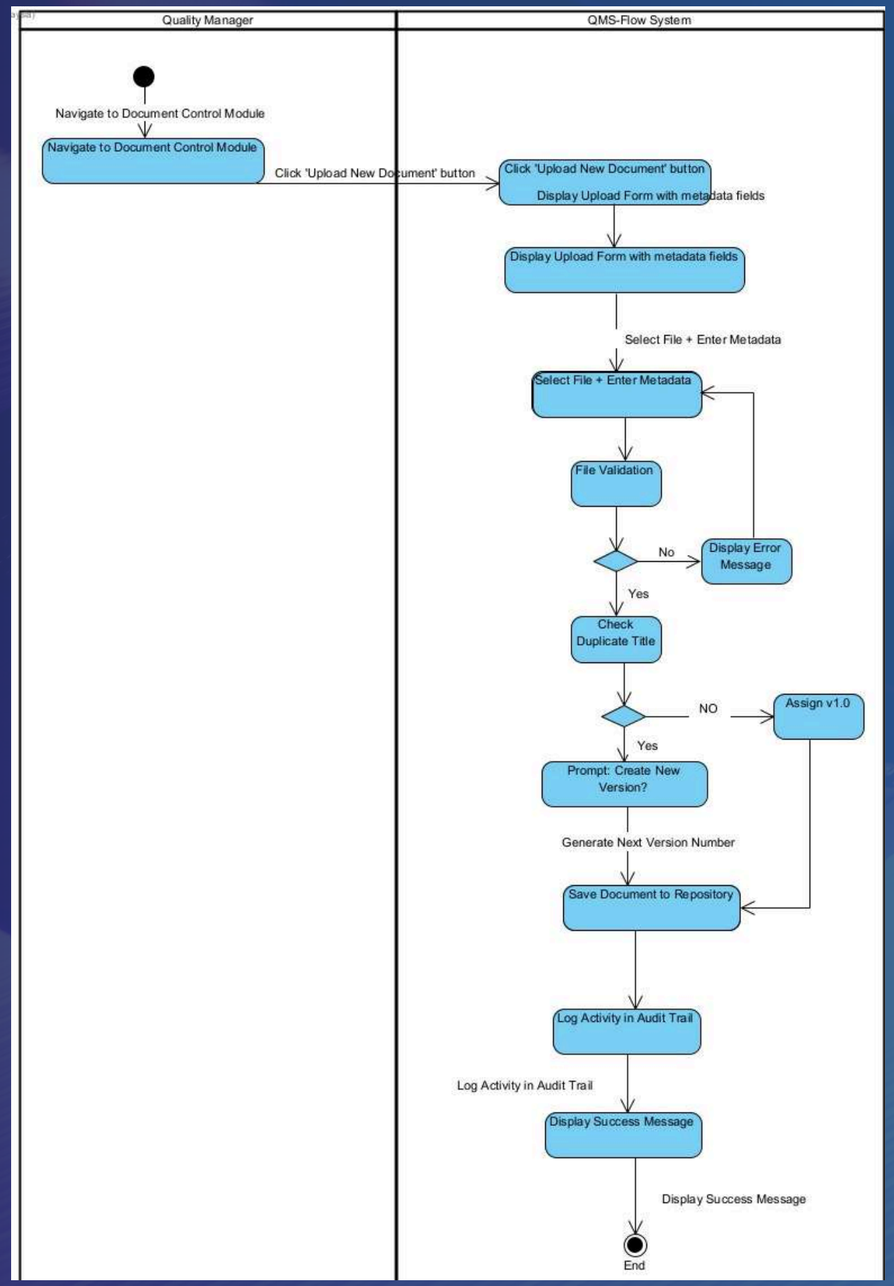

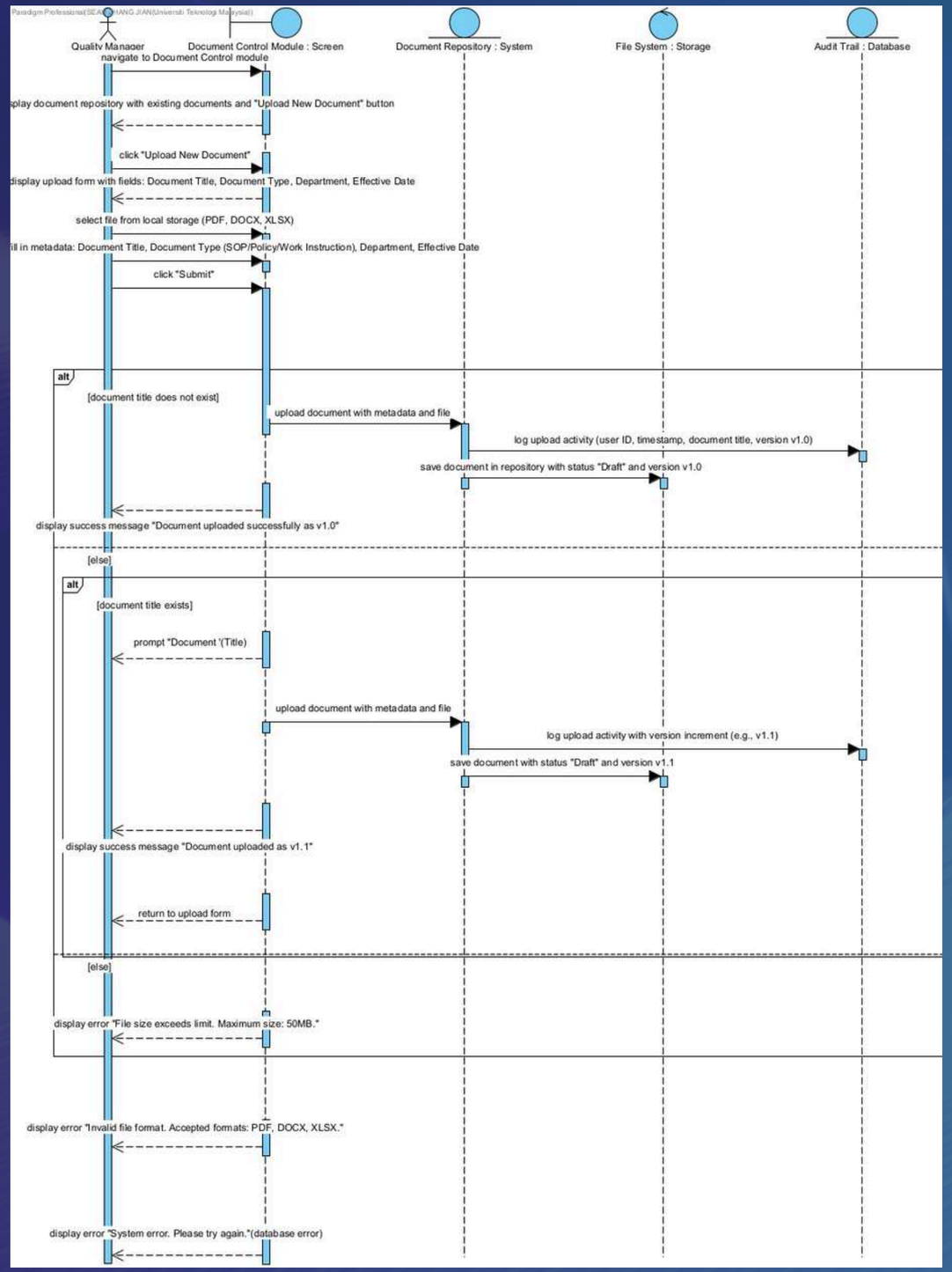
Paradigm Professional(SEAN JIANG JIAN(Universiti Teknologi Malaysia))
Quality Manager
Document Control Module : Screen
Document Repository : System
File System : Storage
Audit Trail : Database
navigate to Document Control module
splay document repository with existing documents and "Upload New Document" button
click "Upload New Document"
display upload form with fields: Document Title, Document Type, Department, Effective Date
select file from local storage (PDF, DOCX, XLSX)
ill in metadata: Document Title, Document Type (SOP/Policy/Work Instruction), Department, Effective Date
click "Submit"
alt
[document title does not exist]
upload document with metadata and file
log upload activity (user ID, timestamp, document title, version v1.0)
save document in repository with status "Draft" and version v1.0
display success message "Document uploaded successfully as v1.0"
[else]
alt
[document title exists]
prompt "Document '(Title)
upload document with metadata and file
log upload activity with version increment (e.g., v1.1)
save document with status "Draft" and version v1.1
display success message "Document uploaded as v1.1"
return to upload form
[else]
display error "File size exceeds limit. Maximum size: 50MB."
display error "Invalid file format. Accepted formats: PDF, DOCX, XLSX."
display error "System error. Please try again."(database error)

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User story: Log Complaint (UC05)

ID: US005

User Story Description
As a Employee, I want to submit the complaint through a simple form so that I can report issues quickly and accurately.

Preconditions:
1. Employee is authenticated and logged in.
2. The complaint logging module is available.

Flow of events:
1. Employee navigates to Log Complaint page.
2. System displays the complaint submission form.
3. Employee enters required details:
   - Customer name
   - Product information
   - Date of complaint
   - Nature/description of issue
4. Employee optionally attaches supporting documents or photos.
5. Employee submits the complaint.
6. System validates the inputs.
7. System saves the complaint record with a unique ID.
8. Use case ends.

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Alternative flow (A1): Missing Required Information The system detects missing required fields and prompts the employee to complete them before resubmission.
Postconditions: 1. A new complaint record is created with status “New”. 2. Notification is sent to the Quality Manager.
Acceptance Criteria 1. Complaint Form Completeness 2. Required Field Validation 3. Attachment Upload 4. Complaint ID Generation 5. Notification to Quality Manager.
Exception flow (E1): System Error During Submission If the system fails to save the complaint, an error message is shown and the employee must retry later.

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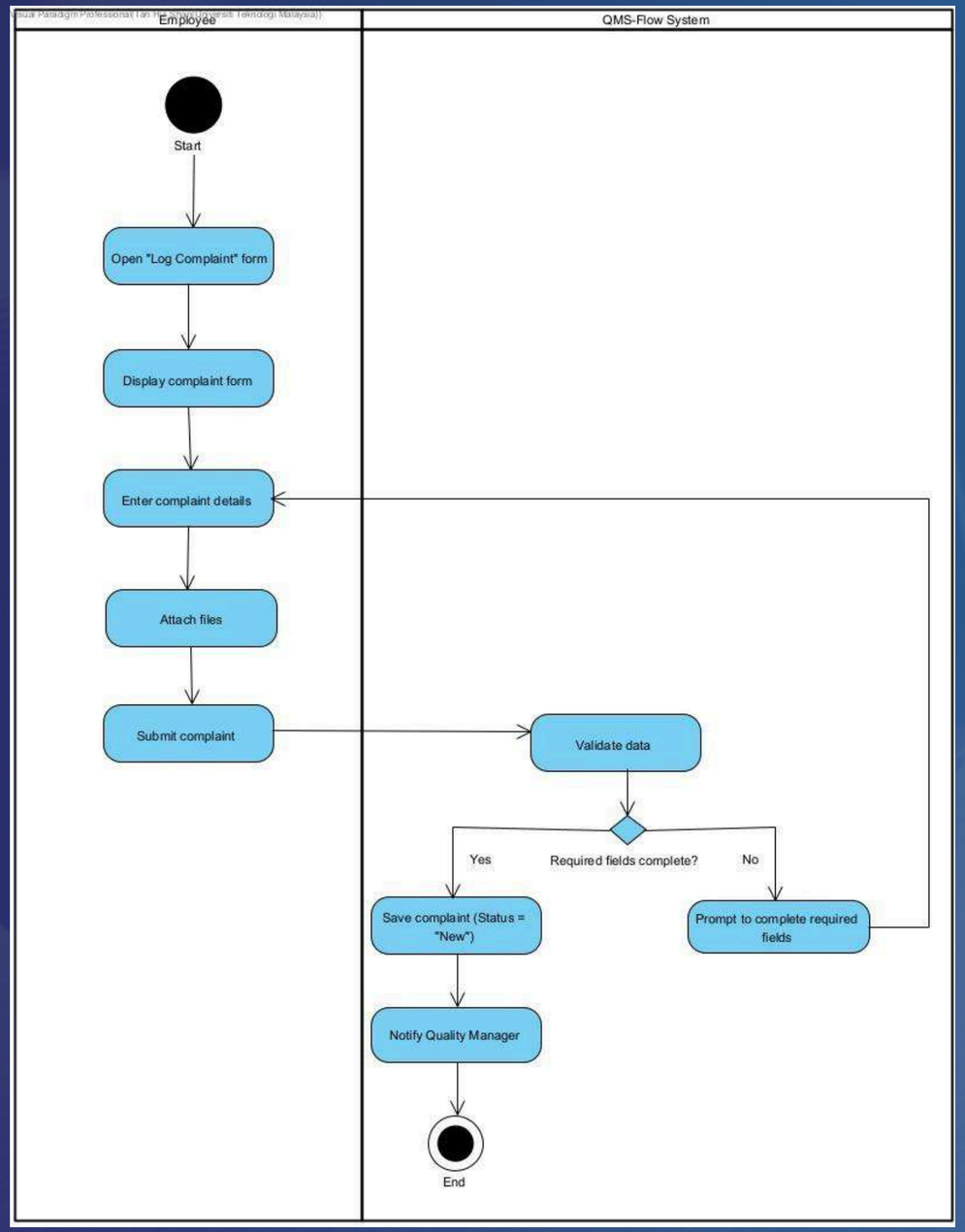

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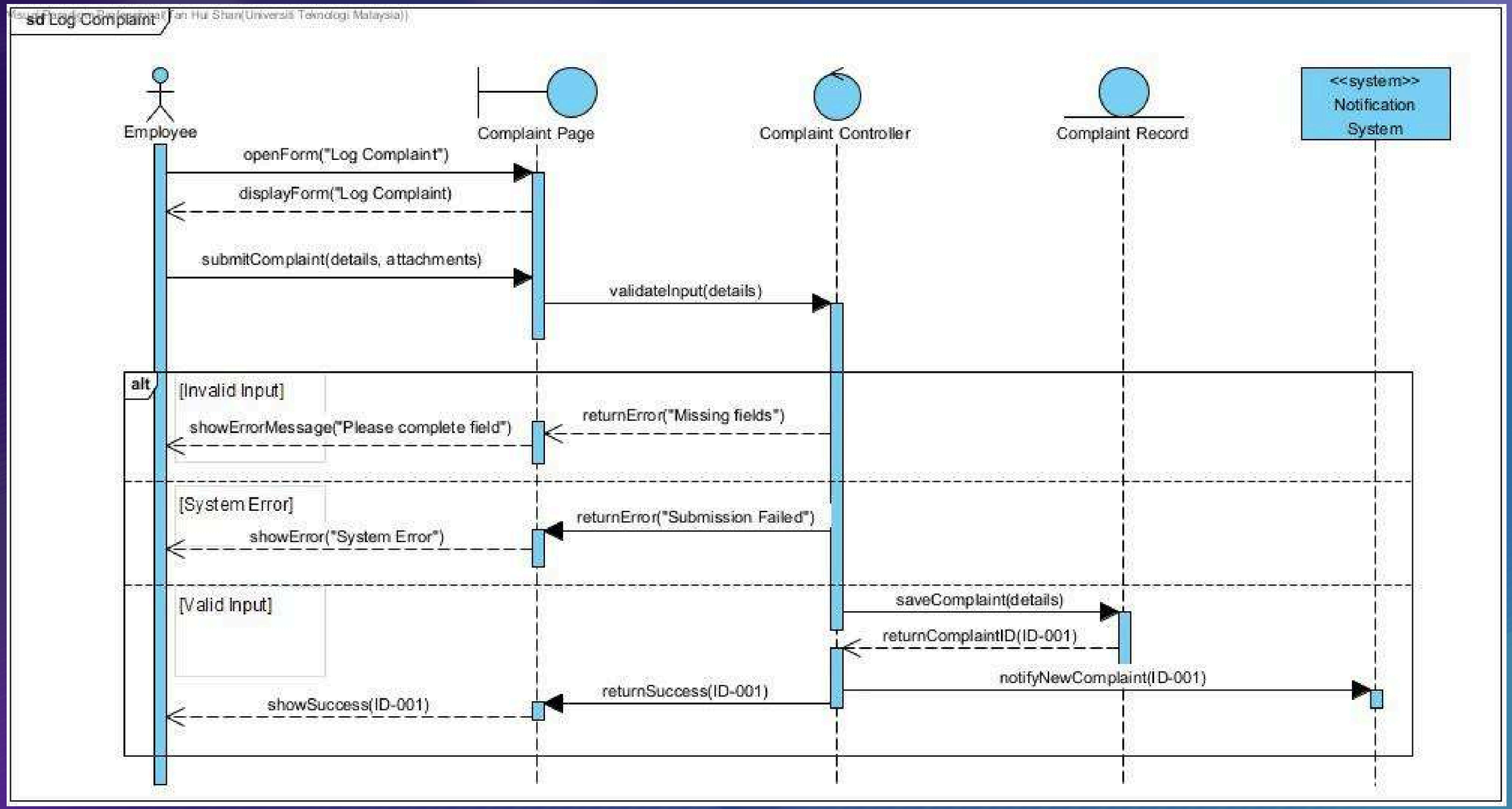

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User story: Initiate CAPA (UC07)

ID: US007

User Story Description
As a Quality Manager, I want to create a formal Corrective Action (CAPA) record linked to a source (Audit Finding or Customer Complaint) so that I can formally document the root cause and assign a corrective action plan.

Preconditions:
1. Quality Manager is authenticated and logged in.
2. A valid source record (Finding or Complaint) exists with a status requiring escalation.

Flow of events:
1. Quality Manager navigates to the Initiate CAPA form.
2. System prompts QM to enter the Source ID (Finding ID or Complaint ID).
3. QM enters the Root Cause Analysis (RCA) and Action Plan details.
4. QM sets the Deadline and assigns the Owner for the corrective actions.
5. QM confirms and submits the CAPA record.
6. System validates the inputs and saves the CAPA record with a unique ID.
7. System updates the status of the source record (Complaint/Finding) to 'CAPA Initiated'.
8. Use case ends.

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Alternative flow (A1): Invalid Source ID
If the Source ID entered (Step 2) is not found, the system displays an error and requires the QM to enter a valid ID before proceeding.

Alternative Flow (A2): Missing Root Cause
The system detects that the Root Cause field is empty (Step 6) and prevents submission, prompting the QM to fill in the mandatory analysis.

Postconditions:

1. A new CAPA record is created with status "Planned".
2. The source record (Complaint/Finding) status is updated to reflect linkage to the CAPA.
3. Notification is sent to the assigned CAPA Owner (Department Head or relevant staff).

Acceptance Criteria

1. CAPA record must successfully link to a unique Finding or Complaint ID.
2. System must validate that Root Cause and Action Plan fields are completed.
3. Assigned Owner must receive an automated notification immediately upon creation.

Exception flow (E1): Database Error during CAPA Creation
If the system fails to save the new CAPA record or update the source record status due to a database integrity error or connection failure, the transaction is rolled back, an error message is displayed, and the Quality Manager must retry the submission.

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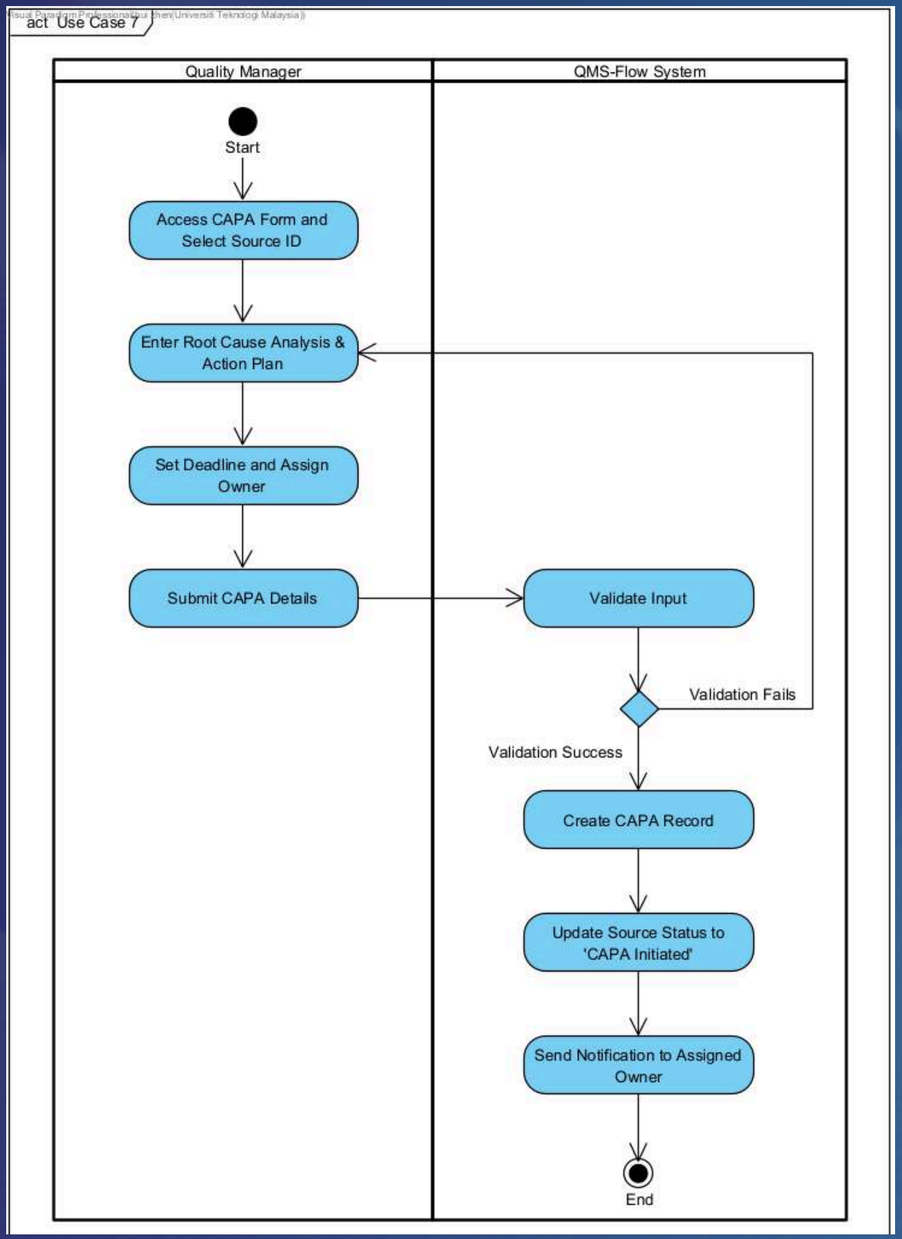

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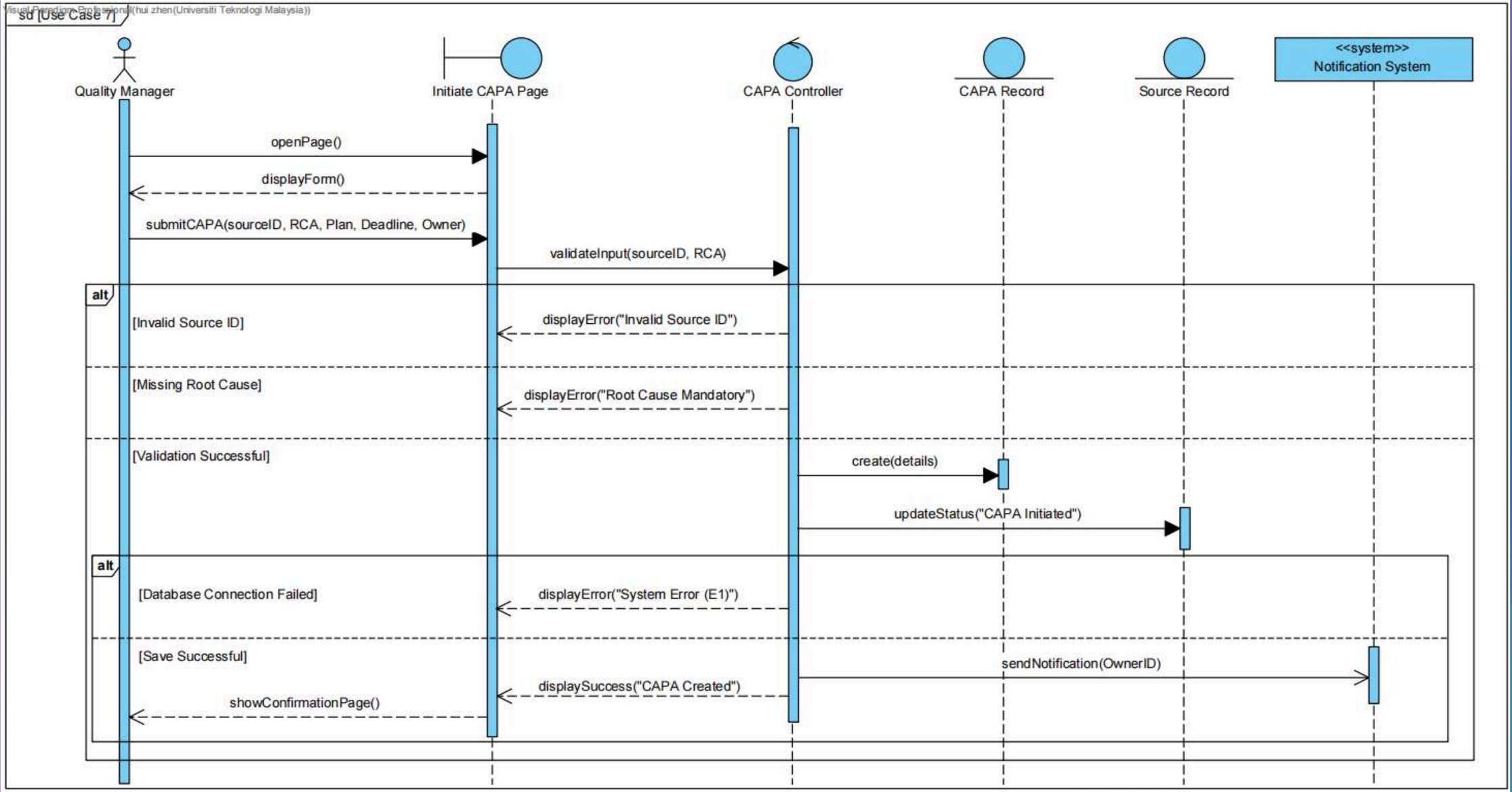

THANK YOU

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MEMBERS:
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TAN HUI SHAN
SEAH ZHANG JIAN
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